Competency Portfolio

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Chapter 1

Part 1: Professionalisation Competency

1.1 Learning Strategy and Approach

The context of this internship made professional self-direction something I had no choice but to take seriously. Zdravoletie is a health and rehabilitation clinic; its staff are trained physiotherapists, fitness professionals, and wellness advisors, not data scientists. The company supervisor, Kostadin Galchin, could speak authoritatively to the clinical purpose of the Anovator bodyAge score and to how practitioners use it in advising clients, but the entire technical direction of the project, including the framing of the research question, the design of the experimental protocol, and the selection of algorithms and evaluation criteria, was necessarily self-determined. There was no in-house technical mentor, no data science team to consult, and no established analytical framework to inherit. Working within this environment required treating professional autonomy not as a privilege to enjoy but as a methodological responsibility: every design choice made without peer review had to be documented carefully and defensible on its own terms.

The strategy adopted in response was to use documentation as a substitute for the external scrutiny that a technical team would ordinarily provide. Every significant decision was recorded in the weekly logbook with the reasoning behind it at the time it was made: the choice of the age gap as the prediction target, the GroupShuffleSplit grouping strategy, the selection of GMM synthesis over generative adversarial alternatives, and the application of Bonferroni rather than less conservative multiple comparison corrections. The record served two functions: it created an audit trail that Peter P.A.L. van Alem could review during supervision, and it forced each decision to be articulated explicitly rather than left as an implicit assumption embedded in code.

Communication with Galchin required a consistent translation effort. Progress updates were provided in Bulgarian to eliminate the language barrier, and the framing was shifted from technical findings to practical implications: not which algorithm achieved the lowest mean absolute error, but what the surrogate model could tell a practitioner about which biometric measurements most strongly predict a client's bodyAge deviation. The Streamlit dashboard was designed with Galchin's perspective as the primary constraint; every visualisation and label is oriented toward clinical interpretation rather than statistical detail. Adapting communication in this way was itself a learning process. Early updates over-indexed on methodological detail; by mid-project, the updates had been restructured to lead with clinical implications and relegate technical qualifications to a separate section for those who wanted them. This approach aligns with the HBO graduate profile competency of professional self-direction, which requires graduates to take responsibility for their own learning process and work autonomously in professional contexts without direct supervision.

1.2 Flexibility in Stakeholder Interactions: Three Moments

Beginning of the internship: initial scope and target variable

The internship began through a mutual connection who identified an alignment between my background in data analysis and Zdravoletie's need for independent research on their Anovator biometric data. The initial meeting with Kostadin Galchin was exploratory rather than directive; he had no specific technical request, only a broader goal: to use the data he had collected to give clients better, evidence-based advice on longevity and health risk. Early discussions ranged widely, including the possibility of incorporating publicly available Bulgarian national health data, but we converged on a more focused scope appropriate to the graduation internship constraints. The first significant decision I proposed was to shift the modelling target away from the Anovator hardware Score, which initial analysis showed to be effectively constant across 85% of clients, toward the age gap (the difference between bodyAge and chronological age). I explained this to Galchin not in statistical terms but in practical ones: the Score told consultants almost nothing useful, while the age gap offered a continuous, sensitive measure that could actually differentiate between clients and track progress over time. Galchin accepted this framing and from that point confirmed full autonomy on the technical direction, with his role being to validate that the research stayed aligned with what would genuinely benefit the business. That working pattern defined the rest of the internship: independent analytical decision-making on my side, accessible communication and directional confirmation on his.

Middle of the internship: midterm review with Peter P.A.L. van Alem

The midterm review, delivered in the first week of April, was the most professionally demanding interaction of the project. At that point the project was structured around a biological age prediction framing and reporting an R^2 of 0.98 as its headline result, both of which Peter P.A.L. van Alem identified as serious problems. The biological age framing was factually incorrect: the study had access to no clinical ground truth of the kind that biological age research requires, and presenting the Anovator bodyAge score as equivalent to a clinically validated biological age measure would expose any performance claim to immediate methodological challenge. The R^2 of 0.98 was implausible given the dataset size and the known noise in the target variable, and indicated a methodological error rather than a strong result. Receiving this feedback required a specific kind of flexibility: the inclination to defend work already completed had to be set aside quickly in favour of a diagnostic response. The immediate decision was to halt new development and trace the source of the artefact before doing anything else. That investigation identified circular benchmarking as the source of the inflated R^2, a structural design error in which synthetic records pseudo-labelled by a model trained on all real data were then used to evaluate that same model. The framing correction and the full eleven-bug audit that followed were direct products of treating the midterm feedback as a methodological signal rather than a criticism to be managed.

End of the internship: closing handover with Galchin

The closing handover took place via message exchange with Kostadin Galchin in the final week of the internship. The Streamlit application was shared with Galchin along with a summary of what the tool does, its known limitations, and an explicit question about whether he found it genuinely useful in practice for advising clients. His response confirmed that the tool is useful and that he sees potential for further development, with a follow-up discussion planned. This was the outcome the communication work throughout the project was aimed at: a non-technical stakeholder independently concluding that an analytically complex tool has practical value. The framing developed over the project — starting from clinical implications rather than technical detail, writing for a practitioner audience, structuring the About page around limitations rather than capabilities — produced a tool that Galchin could evaluate on his own terms and find useful without requiring a technical explanation of how it works.

1.3 Communication Forms Used

Five distinct communication forms were used across the project. Each was chosen to match the audience, the purpose, and the constraints of the professional relationship it served.

Bulgarian-language progress documents (audience: Kostadin Galchin). Written updates in Bulgarian were provided to Galchin at regular intervals throughout the project. The choice of Bulgarian was deliberate and practical: Galchin is a fluent Bulgarian speaker, and conducting all communication in English would have introduced an unnecessary barrier to clear understanding of the project's clinical relevance. These documents were structured around practical implications rather than methodological detail; each update described what the current analysis could say about which biometric measurements drive the bodyAge score and what that might mean for advising practice. Technical qualifications and statistical caveats were present but subordinate to the clinical framing. The form was appropriate because Galchin needed to stay informed of progress and validate that the project remained practically relevant to the clinic's needs, not to evaluate the statistical choices.

Interim report in English (audience: Peter P.A.L. van Alem). The interim report, submitted at the end of Week 8, was the primary communication with the educational supervisor before the midterm. It was written in English in academic register and structured to demonstrate methodological rigour: it covered the research question, the data collection procedure, the feature engineering rationale, and the preliminary results. The form was required by the Fontys assessment structure and appropriate for an academic audience that needed to evaluate the soundness of the research design. The report was the instrument through which Peter P.A.L. van Alem identified the framing and methodology issues that were subsequently corrected.

Midterm oral presentation (audience: Peter P.A.L. van Alem). The midterm presentation was delivered in person to Peter P.A.L. van Alem in the first week of April. The oral form was appropriate because it enabled a live exchange: the presentation of findings could be immediately followed by structured feedback, and the dialogue allowed the framing and R^2 concerns to be communicated directly rather than through written comments on a document. The interactive format made it possible to clarify the source of the concerns in real time and agree on the direction for the corrective work.

Streamlit dashboard (audience: Zdravoletie practitioners; secondary audience: academic assessors). The Streamlit application is a communication artefact as much as a technical one. It translates the analytical outputs of the research, including surrogate model predictions, SHAP feature importance, and population comparison statistics, into a form accessible to a clinical audience with no data science background. The form was chosen because Galchin needed a working tool, not a report, and because a deployed application demonstrating practical usability was a more convincing deliverable than a static visualisation document. The About page, written in plain language and addressed directly to a health professional audience, reinforces the communication function of the tool. The application is also publicly accessible, which allowed it to serve as evidence of product delivery in discussions with both supervisors.

Written thesis (audience: academic examining committee). The graduation report is the primary formal communication of the project's scientific contribution. It is written in English in full academic prose, structured according to the conventions of applied machine learning research: introduction establishing the research question and framing, related work situating the study in the literature, methodology documenting the experimental design and protocol, results reporting all findings with confidence intervals and significance test outcomes, discussion interpreting the findings and their limitations, and conclusion summarising the answers to the sub-questions and identifying future work. The form was appropriate because the audience, Peter P.A.L. van Alem and the examining committee at Fontys, required a complete and verifiable account of the research, not a summary or a practical tool.

1.4 Growth Through Reflection: Three Reflection Reports

Reflection Report 1: February 2026 (beginning of internship)

At the start of the internship I was confident in my technical, analytical, and mathematical abilities. The modelling work itself did not feel intimidating; I had a solid foundation in machine learning and statistics and trusted my ability to work through technical problems independently. The uncertainty lay elsewhere.

Professionally, I was aware of gaps I had not yet fully addressed. Communication had never been a natural strength; I tended to present technical work in technical terms without translating it into what it meant for the person I was talking to. Reliability and consistency in a professional context, rather than an academic one, were also areas I knew needed development. The prospect of working without a technical supervisor felt like both a relief and a risk at the same time. On one side, there would be no pressure to present, explain, and justify every decision constantly.

On the other, those pressures are exactly what develop professional skills, and avoiding them meant I would have to create that pressure for myself.

I did not have a clear research question going in. Peter P.A.L. van Alem had advised that this was normal; research directions shift once you engage with the real data and the real constraints. I accepted that, though I had some quiet concern about whether the project scope would be substantial enough for a graduation thesis. I committed to staying consistent and letting the work develop, while acknowledging that the professional side of the internship would require as much effort as the technical side.

Reflection Report 2: April 2026 (after midterm feedback)

The midterm feedback confronted me with a gap between the confidence I had developed in the preceding weeks and the actual rigour of the work I had produced. I had become increasingly convinced that the $R^2 = 0.98$ result was a genuine finding and had begun to think of the project as technically strong. Peter P.A.L. van Alem's feedback at the midterm was precise and uncomfortable: the framing was wrong, the result was an artefact, and the two problems were connected. My immediate reaction was to want to contextualise and qualify rather than to fully accept the critique; I recall thinking that the result was perhaps inflated but not entirely without signal, and that the framing issue was more a matter of presentation than substance. Both of those instincts were wrong, and working through the audit over the following two weeks confirmed that they were wrong. The R^2 was entirely artefactual. The framing correction was not cosmetic; it changed what claims the thesis was permitted to make. What I learned from this process was less about the specific bugs than about the cost of self-serving interpretation of early results. I had read the 0.98 figure as confirmation rather than as a prompt for scrutiny. Going forward I changed how I read my own results: the first question on encountering a surprising finding became "what could be wrong here" rather than "how strong is this." That shift is visible in how the discussion chapter treats all three experiments; every result, including the statistically significant ones, is examined for alternative explanations before an interpretation is advanced.

Reflection Report 3: June 2026 (end of internship)

The project ended in a different place than I expected at the start, and not only technically. Looking back at Reflection Report 1, the anxiety I identified then was about the professional side of the work: reliability, communicating upward, working without a technical safety net. Those concerns were well-founded. The midterm was the moment that tested them most directly. What changed is that I now have a concrete example of what the cost of delaying self-critique looks like when unchecked, and a documented record of what correcting it required. That is more useful than simply knowing the tendency exists. The communication objective improved more in some directions than others. Working in Bulgarian with Galchin, and structuring updates around clinical implications, became more natural over the course of the project. The Streamlit dashboard reflects that: every label was written for someone who has never heard of mean absolute error. Academic communication to a technical audience remains harder; the midterm feedback made that gap clear, and while the thesis discussion chapter addresses it more deliberately than the interim report did, I would not claim full competence in that register yet. The closing handover with Galchin took place in the final week of the internship. His response confirmed that the tool is useful in practice and that he sees potential for further development. That outcome is the clearest available evidence that the communication objective was met: a non-technical stakeholder, working independently with the tool, found it practically valuable without needing a statistical explanation. The gap that remains — communicating uncertainty in health contexts and building domain knowledge in physiology — is real and is documented in Section 3.4. But the core objective of translating analytical work into something a clinical practitioner finds useful was achieved.

1.5 Sources for Professionalisation Competency

Schön, D. A. (1983). *The Reflective Practitioner: How Professionals Think in Action.* Basic Books.

Schön's foundational account of reflection-in-action and reflection-on-action, meaning the capacity of practitioners to think critically about their own practice while doing it and after the fact, directly informs the three reflection reports in Section 1.4, particularly the midterm reflection on the project reframing: recognising and correcting a flawed framing in response to expert feedback is a textbook instance of the reflective competence Schön describes.

Nisbet, M. C., & Scheufele, D. A. (2009). What's next for science communication? Promising directions and lingering distractions. *American Journal of Botany*, 96(10), 1767–1778 (published in a special issue on science communication). `https://doi.org/10.3732/ajb.0900041`

This paper analyses the challenge of translating scientific findings for non-specialist audiences and identifies stakeholder-oriented framing as a key strategy, directly applicable to the challenge of communicating a machine learning research project to a clinical company supervisor with no statistical background.

Chapter 2

Part 2: Learning Objective

2.1 The Learning Objective

The learning objective for this internship is to strengthen my ability to communicate analytical results clearly and effectively to non-technical stakeholders and to translate data-driven insights into actionable recommendations.

This will be achieved by regularly presenting intermediate results to the company supervisor, discussing assumptions, limitations, and implications of the models, and adapting my communication style based on stakeholder feedback. Active participation in feedback discussions will be used to improve clarity, alignment, and professional interaction.

Progress will be measured by the quality and consistency of stakeholder feedback, the level of alignment between analytical outcomes and stakeholder expectations, and the ability to independently lead result discussions and decision-oriented conversations.

2.2 Reason for This Learning Objective

The learning objective was chosen in recognition of a professional gap that this particular project was well positioned to expose. Technical analytical work, including designing experiments, running models, and interpreting statistical outputs, takes place in a constrained and legible environment where quality criteria are relatively clear: results are reproducible or they are not, methods are appropriate to the data or they are not, claims are supported by evidence or they are not. Communicating that work to stakeholders who do not share the same technical vocabulary is a different kind of challenge, and one that a data scientist working in a professional advisory context cannot avoid.

The specific structure of this internship made this gap especially salient. Kostadin Galchin, the company supervisor, is a health and fitness professional who uses the Anovator system daily and has direct opinions about what the bodyAge score means to clients and practitioners. He does not, however, have a background in statistical modelling. Conveying to him what it means for a Wilcoxon test comparison to be inconclusive, or why a mean absolute error of 2.43 years is simultaneously the best available result and an honest acknowledgement of the limits of what 53 individuals can establish, required a translation effort easy to underestimate. The same challenge applied to the Streamlit dashboard: every design decision about what to show, how to label it, and what to leave out was a communication decision as much as a technical one. The learning objective was chosen because this internship presented a concentrated version of the challenge that defines professional data science work at the boundary between analysis and decision-making: being useful to people who need your results but cannot evaluate your methods.

2.3 How the Learning Objective Was Worked On

Progress toward the learning objective was made through a set of concrete actions that spanned the full project period. The earliest and most sustained action was the practice of writing regular progress updates in Bulgarian for Galchin. Each update required explicitly asking: what does this finding mean for a practitioner advising clients on diet and exercise, and how can I explain it without requiring the reader to have any statistical background? The question had no obvious answer at the start of the project. Trying to answer it repeatedly, watching whether Galchin's responses indicated it had landed, was the main vehicle for improvement.

The midterm presentation was a second, more concentrated opportunity. Preparing a twenty-minute oral account of the project for Peter P.A.L. van Alem forced a decision about which elements were essential and which were internal scaffolding that the audience did not need. The feedback that the biological age framing was wrong was, in retrospect, partly a communication failure: the framing had been adopted without sufficiently interrogating whether it was defensible to a careful academic listener, not just familiar within the project.

Building the Streamlit dashboard shifted the communication work into a design register. The constraints were tighter there; every label, tooltip, and page title either communicated something useful to a health professional or it did not, and there was no prose available to compensate for a poorly chosen visualisation. Writing the About page required a plain-language account of the tool's limitations that could be understood by someone who had never heard of mean absolute error. Completing that page was one of the clearest tests of whether the learning objective had been achieved in practice.

2.4 Scale Question: Three Measurements

The scale question measures the degree to which the student can effectively communicate technical analytical findings to non-technical stakeholders, on a scale of 1 (unable to adapt communication to audience) to 10 (consistently communicates complex findings clearly and accessibly across all audience types).

Start of internship: February 2026

At the start of the internship I would honestly rate myself a 3 out of 10 on this scale. I had no significant experience presenting analytical findings to non-technical decision-makers, and my previous communication in academic settings had been largely peer-to-peer, with classmates and fellow students where I felt genuinely confident. The specific challenge of communicating upward, to supervisors, managers, or stakeholders, was one I consistently struggled with. In those situations I became overly focused on making a good impression rather than communicating clearly, which had the opposite effect; the anxiety about being judged interfered with the quality of the explanation itself. I was aware of this pattern going into the internship and recognised it as something that needed to change. The Zdravoletie context, a non-technical supervisor who needed accessible, actionable insights rather than statistical detail, was precisely the kind of environment that would force me to address this directly. That awareness was the starting point, but awareness alone does not translate to ability, which is why I rate myself a 3 rather than higher.

Middle of internship: April 2026

5/10. The midterm review with Peter P.A.L. van Alem provided direct evidence of progress and its limits simultaneously. On the progress side, the project's practical framing, which involved working in Bulgarian with a non-technical supervisor and structuring updates around clinical implications, had required consistent communication adaptation throughout the preceding weeks, and that practice had produced measurable improvement in how clearly progress could be explained to Galchin. The feedback from Peter P.A.L. van Alem indicated, however, that the same improvement had not yet carried into the academic communication register. The midterm presentation and the interim report both over-indexed on technical detail and under-indexed on the implications of the findings: what do these results mean for a practitioner, why does the framing matter for how conclusions are drawn, and what can and cannot be claimed from the evidence available. The 5/10 rating reflects genuine progress in non-technical stakeholder communication but persistent limitation in a second area: disciplined, implication-focused academic communication to a sophisticated technical audience.

End of internship: June 2026

7 out of 10. Progress was genuine in one direction and limited in another. Communication with a non-technical stakeholder in an applied setting improved substantially: structuring updates around implications, writing for a clinical audience, building a tool that explains itself without statistical jargon. The Bulgarian-language updates, the dashboard design, and the About page are concrete evidence of that. The two concrete deliverables that account for the gain from 5 to 7 are the completed Streamlit dashboard, which required every communication decision to be made for a clinical audience with no statistical background, and the thesis discussion chapter, which addresses implication-focused academic communication more deliberately than the interim report did. The gap that remains is in the academic register: presenting findings to a technically sophisticated audience with the right balance of precision and implication. The midterm feedback identified this clearly. I would not yet claim full competence in that register. The 7 rating reflects genuine progress from 3 at the start and 5 at midterm, with the honest acknowledgement that the remaining gap is real. The most meaningful external assessment of this objective comes from two sources: Kostadin Galchin's confirmation that the tool is useful in practice for advising clients and that he sees potential for further development, and the educational supervisor's evaluation

of the thesis and this portfolio. Galchin's response is direct evidence that the communication objective was achieved in the applied, non-technical stakeholder direction.

2.5 Feedback and Feedforward from Four People

Kostadin Galchin is the company supervisor at Zdravoletie Health and Rehabilitation Centre in Burgas. He is a health and fitness professional with daily operational experience using the Anovator system and has no background in statistical modelling or machine learning.

Feedback (original, Bulgarian):

> Забелязвам проявеният интерес и вникването в детайли относно новата материя. Правилното разбиране ти е позволило да се справиш толкова добре със заданието. Работният процес минава гладко, професионално, също като комуникацията помежду ни. Чрез този стаж определено си развил и демонстрирал умения, придобити през учебния процес години назад. Прогресът е видим и тепърва ще го усъвършенстваш, респективно си готов за по-обширни и сложни проекти.

Feedback (English translation):

> "I notice the interest shown and the attention to detail regarding the new subject matter. Properly understanding it has allowed you to handle the assignment so well. The working process has gone smoothly, professionally, as has the communication between us. Through this internship you have clearly developed and demonstrated skills acquired during your studies years ago. Progress is visible and you will keep refining it further; accordingly, you are ready for broader and more complex projects."

Feedforward (original, Bulgarian):

> Да се захващаш с нови материи и ниши, които ще са актуални години напред – това е инвестиция в бъдещето ни. Да обогатяваш всячески общата си култура – ключът към прогреса! Винаги да търсиш добавената стойност в личния живот и работата си – да бъдеш в помощ на човечеството, да решаваш проблеми на близки и приятели.

Feedforward (English translation):

> "Take on new subjects and niches that will remain relevant years ahead – this is an investment in our future. Enrich your general knowledge in every way possible – the key to progress! Always look for added value in both your personal life and your work – be of help to humanity, solve problems for those close to you and for friends."

Response: Galchin's feedback affirms what was most valuable about the working structure of this project: being trusted to set the direction and make technical decisions independently created a level of ownership that would not have been possible in a more supervised environment. The self-directed nature of the internship forced me to develop analytical accountability. Every design choice was made by one person and had to be defensible on its own terms, without the safety net of a technical peer review process. The feedforward on engaging with new fields resonates directly with what this project demonstrated. The ability to develop functional competence in an unfamiliar domain, such as health informatics and clinical score modelling, is a form of professional capital that compounds over time. I intend to treat breadth of domain knowledge as a deliberate investment rather than an incidental byproduct of whatever project arrives next.

Dani is a classmate from Fontys University of Applied Sciences with whom I have worked across multiple academic projects and courses over three years. He has observed my technical and professional development directly in collaborative academic and project settings.

Feedback (verbatim):

> "From experience with Alexander on several projects, he is quite adept at taking on hard tasks and meeting deadlines."

Feedforward (verbatim):

> "He could however improve showing how impressive his work truly is. This could be done by updating his supervisors frequently on his work. If this is achieved, he will for sure gain that mass respect."

Response: This feedforward identifies a pattern I recognise. My tendency is to work independently until something is finished before surfacing it, which means the process behind a result is often invisible to the people who need to see it. In the context of this internship, the communication learning objective was specifically about addressing this: structuring regular updates for Galchin, presenting interim findings to Peter P.A.L. van Alem, and building a tool that explains its own outputs. The feedforward confirms that visibility of effort is a distinct skill from quality of effort, and one I need to develop more deliberately going forward.

Daniel is a professional colleague from a collaborative project context at Accedia. He has strong experience in machine learning engineering and software development, and the collaboration involved building a recommendation system.

Feedback (verbatim):

> "Aleksandar has been part of the Data Science team for the past 3 months as of May 2026. Since the beginning he showed very strong foundation of machine learning and Python concepts. He has been using his theoretical and practical knowledge to tackle important and challenging tasks while building a recommendation system. Highlights of his effort are the quality of his work and the well-documented outcomes of the tasks."

Feedforward (verbatim):

> "The skill that would improve himself as a data scientist and a developer as a whole in the long-term is to not be afraid to ask for assistance after spending time on a problem without reaching the desired outcome. Technical wise the software world

has been going to the cloud for years now so learning about the products of the big cloud providers is always a good path."

Response: The feedforward on asking for help after sustained effort resonates as a fair and accurate observation. My default pattern when blocked is to extend the independent effort by running one more experiment, consulting one more paper, or restructuring the problem, before surfacing the difficulty to a colleague. There were moments in this project where that instinct extended the timeline unnecessarily rather than serving the work. I recognise this as a genuine development priority rather than a minor stylistic preference, and I intend to treat knowing when to escalate as a deliberate skill to practice in collaborative settings going forward. The cloud provider feedforward is one I accept unreservedly. The trajectory of professional data science work in the next several years will be substantially shaped by cloud infrastructure competency, and the gap between what I understand analytically and what I can deploy at scale in a cloud environment is one I plan to close through structured study and practice.

Alexander Tanev's father is an observer from the home and family environment, without professional involvement in the project.

> *"As a father who observes Alexander closely at home, I can say that since the start of his internship I have noticed a visible change in his maturity and sense of responsibility. He approaches his work seriously and I can see how much effort he puts in, often working late into the evening on his project. If I were to give one piece of advice for his development, it would be to act more quickly and decisively when a situation requires it; he sometimes spends a long time thinking before taking action, and building confidence in his own decisions will serve him well going forward."*

— Alexander's father, May 2026

Response: The observation about decisiveness is accurate, and I recognise it without qualification. The clearest illustration from this project is the three weeks I spent after first obtaining the $R^2 = 0.98$ result before the midterm. I had a persistent unease about that figure but continued building on it rather than stopping immediately to investigate, extending both the time wasted and the scope of the necessary correction when the problem was eventually identified. A more decisive response to early doubt, halting development and auditing the data flow on the day the suspicion first arose, would have resolved the issue in hours rather than weeks. That pattern of extended deliberation before acting on a clear signal is one I intend to address directly. The practical change I am working on is to set explicit time limits for uncertainty before committing to a diagnostic action. Reflection serves decision-making rather than postponing it.

2.6 Sources for Learning Objective

Spiegelhalter, D., Pearson, M., & Short, I. (2011). Visualizing uncertainty about the future. *Science*, 333(6048), 1393–1400. `https://doi.org/10.1126/science.1191181`

Spiegelhalter and colleagues analyse how uncertainty in quantitative predictions can and should be communicated to non-specialist audiences, presenting evidence-based recommendations for displaying confidence intervals and probability distributions, directly applicable to the challenge of communicating the model's 95% CI results and 2.43-year MAE to Zdravoletie practitioners who need to act on these outputs without a statistical background.

Davenport, T. H., & Patil, D. J. (2012). Data scientist: The sexiest job of the 21st century. *Harvard Business Review*, 90(10), 70–76.

Davenport and Patil describe the data scientist's role as fundamentally advisory, translating analytical findings into organisational decisions by working at the boundary between technical analysis and non-technical stakeholders, and their characterisation of the required communication competencies maps directly onto the learning objective selected for this portfolio: the challenge of making complex model outputs accessible to a non-technical company supervisor and clinical end users.

Chapter 3

Part 3: Overall Reflection

3.1 Approach and Decision-Making

Four decisions shaped the character of this project, and reflecting on each reveals something about how analytical choices interact with professional integrity in research practice.

The first was choosing the age gap (bodyAge minus chronological age) as the prediction target rather than the raw bodyAge score. The reasoning was direct: chronological age is trivially predictable, and including it as a component of the target variable would inflate performance metrics without illuminating the question of interest, which is what the Anovator formula does with the physiological deviation from expected age. The decision was made at the start of Week 1 and logged in the logbook. Alternatives considered were predicting bodyAge directly and predicting a binary indicator of elevated bodyAge. Both were rejected: the first for the reason stated above, the second because it discards continuous variation that the regression framework can exploit. The decision was straightforward technically but important to document; a reader who did not understand it might misinterpret the model's mean absolute error of 2.43 years as a direct error on bodyAge itself.

The second was choosing GMM synthesis over CTGAN, TVAE, or copula alternatives. The reasoning was data-scale: 158 records falls below the regime in which GAN architectures train reliably, and the instability of GAN training at small data scales would make results from such a comparison unreliable and hard to interpret. The GMM was not the most powerful generative approach available; it was the most appropriate one given the constraint. What this decision taught me is that in applied research, the right method is not the most sophisticated method but the most honest method for the available data. Choosing a simpler approach because it is defensible is preferable to choosing a complex approach because it looks more advanced.

The third was applying Bonferroni correction rather than a less conservative multiple comparisons procedure. The Benjamini-Hochberg procedure, for example, would have produced more significant results across all three experiments. The decision to apply Bonferroni reflected a prior about the primary risk in this context: a false positive claim that one algorithm or training regime significantly outperforms another, when the data cannot actually certify that, would misrepresent the project's contribution. The consequence was that the majority of comparisons were reported as inconclusive, which was harder to present but more honest.

The fourth decision, reframing the project after the midterm, was the most difficult and ultimately the most formative. The original biological age framing was not merely a presentation choice; it was a substantive claim that the Anovator score constituted clinical ground truth of the kind studied in the biological age literature. Abandoning it required accepting that weeks of work had been built on an incorrect premise. The reverse-engineering framing was identified as the correct alternative because it made no claims about clinical validity: the model approximates what the Anovator formula produces from biometric inputs, not what a physiologically validated biological age measurement would produce, and that distinction is defensible on the data available. The lesson was about the distinction between investing effort in an approach and being correct about that approach, and about the importance of treating expert feedback as evidence rather than as an obstacle.

3.2 Collaboration and Communication

This project was largely self-directed, and that condition had real advantages and real disadvantages worth examining. The advantage was full ownership of every analytical decision: no technical manager to negotiate with, no conflicting priorities to balance, no institutional inertia to overcome. Every choice about experimental design, evaluation protocol, and statistical framing was made by one person, based on explicit reasoning logged at the time. That ownership produced a kind of analytical coherence that is harder to achieve when a research direction is negotiated across a team.

The disadvantage was the absence of peer review for those same decisions. In a team environment, a proposed evaluation protocol would be read and questioned by a colleague before the first line of code was written. The circular benchmarking error that produced the $R^2 = 0.98$ artefact, the most significant methodological mistake in the project, persisted for several weeks precisely because no one else read the data flow across the notebooks and asked why the evaluation set overlapped with the pseudo-label source. Peter P.A.L. van Alem's midterm question, which surfaced the problem, was the external review that should have happened earlier. That experience is the clearest evidence of what peer review actually provides that solo work cannot replicate through documentation alone.

Communication with Galchin improved significantly over the project period. The initial updates were too dense and the clinical implications too buried. By mid-project, the structure had been inverted: implications first, methods as supporting context. The dashboard is the fullest realisation of that approach; it shows the SHAP chart and the risk/protective factor labels before it mentions the model architecture, and the About page addresses a practitioner rather than a researcher. Galchin's feedback confirmed that this translation succeeded: he found the tool genuinely useful in practice and identified potential for further development without needing a statistical explanation of how it works.

3.3 Evidence of Development

The following artefacts support the competency and learning objective claims made in this portfolio. All files are included in the graduation submission package.

Midterm presentation slides (April 2026) (`I_Present.pdf`). The slide deck prepared for the April 22 midterm presentation with Peter P.A.L. van Alem demonstrates the state of technical communication at the midpoint of the internship. The slides show how findings were structured for an academic audience at that stage, including the biological age framing and the $R^2 = 0.98$ result that were later challenged. As a before-and-after reference point alongside the reframed thesis, the presentation illustrates the development in both framing accuracy and communicative clarity across the internship period.

Interim report (March–April 2026) (`I_Internship.pdf`). The interim report represents the state of technical communication at the midpoint of the internship. It is evidence of both capability and limitation: the document shows how to structure a research narrative for an academic audience, but also reflects the biological age framing error and the uncritical presentation of the $R^2 = 0.98$ result. As a before-and-after reference point it is more useful than a document that shows only success.

Midterm feedback form (`Mid-term Feedback Form - 20260422_ALEP.docx` and `Interim_Evaluation_Form_Company_Graduation_Project_A_Tanev_Final.docx`). The written feedback from Peter P.A.L. van Alem following the April 22 midterm presentation. The document records the two critical challenges raised, the incorrect framing and the implausible performance result, and documents the communication and methodological gaps that the second half of the internship was designed to address.

Thesis methodology chapter (`Thesis.pdf`, Chapter 5). The methodology chapter communicates a complex experimental protocol, including the GroupShuffleSplit design rationale, Wilcoxon signed-rank testing with Bonferroni correction, and three-regime synthetic data ablation, with precision and without ambiguity. Compared to the interim report, it reflects a substantially higher standard of academic communication and a more honest treatment of limitations.

Thesis discussion chapter, Section 7.4: SHAP interpretation (`Thesis.pdf`, Section 7.4). This section translates SHAP model output into physiological terms accessible to a health professional audience, explaining what high waist circumference SHAP values mean for this population and connecting feature importance to existing understanding of metabolic health risk. It bridges technical model output and domain-relevant interpretation, which is the applied form of the learning objective.

Streamlit application About page (`0_About.py`). The About page of the deployed Streamlit dashboard was written explicitly for a non-technical audience. It explains what the tool does, what the model predicts (an approximation of Anovator's bodyAge score, not clinical biological age), and what the known limitations are, in plain language without statistical jargon. The deliberate framing of limitations rather than capabilities reflects the communication standard developed over the course of the internship. The live application, including this page, is publicly accessible at `https://zdravoletie-app-r59gvphlkih7549cgipbgt.streamlit.app/About`.

Internship logbook (`Logbook.pdf`). The logbook documents decisions, challenges, and reflections across all nineteen weeks of the internship. It records the systematic documentation practice adopted to compensate for the absence of peer review in a self-directed research environment. The quality of reflection in later entries, compared to the more descriptive early entries, shows development in analytical self-awareness over the internship period.

3.4 Areas for Further Development

Four areas of professional capability were exposed as underdeveloped by the experience of this project and are identified here without qualification as directions for continued work.

The first is the communication of statistical uncertainty to non-statisticians. Reporting a mean absolute error of 2.43 years with a 95% confidence interval of [2.09, 2.76] is precise and defensible in academic context. Explaining to a practitioner what it means that the prediction could be off by more than two years on a given client, and what that implies for the weight they should give the score in an advising conversation, is a different and harder task. The Streamlit dashboard addresses this imperfectly, and the limitation section of the About page is necessarily compressed. A more complete treatment would require fluency in communicating uncertainty in health contexts specifically, a field with its own literature and practice standards that this project touched but did not engage with systematically.

The second is working within structured organisational environments. Zdravoletie provided minimal procedural structure, and the flexibility that resulted was both an advantage and a gap in professional experience. The ability to work within a structured organisation, where analytical decisions must be approved, where communication has defined channels, and where timelines are constrained by organisational processes rather than personal planning, is a capability that this internship did not exercise.

The third is domain knowledge in health and physiology. The project required biological interpretation of SHAP feature importance outputs, explaining for example why trunk fat distribution and bioimpedance phase angle appeared among the top predictors in physiological terms, without a biology or physiology background. The interpretations offered in the thesis are grounded in the research literature but would benefit substantially from direct collaboration with a clinician.

The fourth is the discipline of building conceptual understanding and practical skill independently, without offloading execution to automated tools. Feedback from a colleague during this project period raised a concern that over-reliance on AI-assisted code generation can produce working outputs while bypassing the reasoning process that builds durable professional competence; a graduate-level practitioner who consistently reaches for a tool to produce code or documentation rather than writing it by hand risks accumulating a surface familiarity with techniques without the depth needed to diagnose, adapt, or defend them under pressure. That observation is valid and I take it seriously. The correct response is not to refuse useful tools but to be deliberate about when assistance accelerates legitimate work and when it substitutes for understanding that I should be building myself. Writing more code, derivations, and documentation by hand, and tolerating the slower pace that entails, is a necessary discipline at this stage of professional development, and one I will treat as a deliberate practice rather than an occasional choice.

Appendix A

Ethical Considerations

A.1 Data Classification Under GDPR

The dataset used in this study consists of Anovator body scan records collected from clients of Zdravoletie. Under the General Data Protection Regulation (EU) 2016/679, this data falls into two protected categories. First, the name field constitutes personal data under Article 4(1), as it directly identifies natural persons. Second, the biometric and health measurement fields (including body composition percentages, impedance values, postural risk scores, blood pressure, heart rate, vision acuity, and the derived bodyAge score) constitute special category data under Article 9(1), specifically data concerning health.

Of the 158 records in the dataset, 103 carry the name value Unknown. The remaining 50 records carry values that appear to be dates of birth encoded as six-digit strings, which are pseudonyms rather than true names but remain linkable to individuals within the clinic's operational records. The original_url and id fields contain Anovator platform identifiers that could be used to retrieve additional personal data; these fields were not used in any modelling or analysis and are excluded from the model feature set.

A.2 Consent Assumptions

The data were collected in the course of normal clinical operations at Zdravoletie. It is assumed that Zdravoletie's standard client intake process includes consent to collection and processing of biometric data for health assessment purposes. However, this study did not verify whether clients were specifically informed that their scan data would be used for machine learning research or shared with a student researcher. This represents a gap relative to the Article 9(2)(a) explicit consent standard for research use of special category data.

A.3 Data Handling in This Study

The dataset was transferred to the researcher's local development environment as a CSV export and was not uploaded to any public repository, cloud storage service, or third-party platform during analysis. All modelling, experiment scripts, and visualisations were executed locally. The Streamlit dashboard does not transmit client data to external servers; all prediction and SHAP computation runs client-side within the Streamlit session. Model artifact files (.joblib) stored in the repository contain only fitted model parameters and no training records.

A.4 Requirements for Production Deployment

A production deployment of the Streamlit dashboard or any successor system would require several steps before going live. A Data Protection Impact Assessment (DPIA) under Article 35 GDPR must be completed. Explicit informed consent forms covering research and analytical use of scan data must be in place, alongside a Data Processing Agreement under Article 28 GDPR with any cloud hosting provider. Access must be restricted to authorised clinic staff through authentication, with audit logging of record access. A documented data retention policy is also required for compliance with Article 17 erasure requests.